

hovid Hydrocortisone Sodium Succinate for Injection 100 mg

VICORZC-M3

DESCRIPTION

A white or almost white powder filled in 5 ml amber glass vial.

COMPOSITION

Each vial contains Hydrocortisone Hydrogen Succinate 127.6mg equivalent to Hydrocortisone Sodium Succinate 133.7 mg which equivalent to Hydrocortisone 100mg.

ACTIONS AND PHARMACOLOGY

Glucocorticoids, naturally occurring and synthetic, are adrenocortical steroids that are readily absorbed from the gastrointestinal tract. Naturally occurring glucocorticoids (hydrocortisone and cortisone), which also have salt retaining properties, are used as replacement therapy in adrenocortical deficiency states. Their synthetic analogs are primarily used for their anti-inflammatory effects in disorders of many organ systems. Glucocorticoids cause profound and varied metabolic effects. In addition, they modify the body's immune response to diverse stimuli. Hydrocortisone sodium succinate has the same metabolic and anti-inflammatory actions as hydrocortisone. When given parenterally and in equimolar quantities, the two compounds are equivalent in biologic activity.

PHARMACOKINETICS

The highly water-soluble sodium succinate ester of hydrocortisone permits the immediate intravenous administration of high doses of hydrocortisone in a small volume of diluent and is particularly useful where high blood levels of hydrocortisone are required rapidly. Following the intravenous injection of hydrocortisone sodium succinate, demonstrable effects are evident within one hour and persist for a variable period. Excretion of the administered dose is nearly complete within 12 hours. Thus, if constantly high blood levels are required, injections should be made every 4 to 6 hours. This preparation is also rapidly absorbed when administered intramuscularly and is excreted in a pattern similar to that observed after intravenous injection.

INDICATIONS

Hydrocortisone sodium succinate is indicated in situations requiring a rapid and intense hormonal effect.

Acute Adrenocortical Insufficiency: This syndrome may be induced by severe stress (e.g., surgery, trauma, or infection) in patients with Addison's disease, panhypopituitarism, or latent adreno cortical insufficiency due to corticosteroid therapy. Patients in these categories should be prepared for elective surgery with prophylactic doses of cortisone or hydrocortisone. Should evidence of adrenal insufficiency develop despite preparation, hydrocortisone sodium succinate should be administered promptly to support the patient. Cases in this group which require emergency surgery that does not permit prophylactic preparation with steroids should receive this product intravenously before the operative procedure and at the same time be started on intramuscular cortisone or hydrocortisone. The latter therapy should be continued for an appropriate interval into the postoperative period.

Bilateral Adrenalectomy: Patients who are to undergo this procedure should be prepared with intramuscular injections of cortisone and hydrocortisone before surgery. Hydrocortisone sodium succinate should be given intravenously immediately, prior to operation and at appropriate intervals to support the patient through the period of maximum stress. Steroid dosage should be tapered following the procedure and the patient eventually transferred to oral replacement therapy.

Severe Shock: In severe shock adjunctive use of intravenous hydrocortisone sodium succinate may aid in achieving hemodynamic restoration. Corticoid therapy should not replace standard methods of combating shock, but present evidence indicates that concurrent use of large doses of corticoids with other measures may improve survival rates.

Acute Hypersensitivity Reactions: In status asthmaticus, and allergic drug anaphylactic reactions, epinephrine or other vasopressor substances should be given before or along with hydrocortisone sodium succinate.

Overwhelming Infections with Severe Toxicity: In patients moribund from overwhelming infections for which specific antibiotic therapy is available, intensive hydrocortisone sodium succinate therapy may permit survival until the antibiotic has time to take effect. Necessary procedures for the establishment of a bacterial diagnosis should be carried out and intensive antibiotic treatment begun on the basis of the proven or probable etiology before steroid therapy is started. In the presence of infection, this product should be administered for the shortest time compatible with adequate clinical response and must be discontinued before the antibiotics by at least 3 days. In the case of surgical infections definitive surgical therapy should be scheduled as promptly as the patient's condition permits. Clinical improvement resulting from steroid therapy must not be cause to defer surgical treatment.

Systemic Lupus Erythematosus in Relapse: In this condition intravenous administration of hydrocortisone sodium succinate is of value in initiating therapy. Oral therapy with appropriate doses of adrenal steroids should be employed as soon as clinical improvement occurs.

Aspiration Pneumonitis: Intravenous administration of hydrocortisone has been found to be beneficial in the management of pneumonitis produced by aspiration of vomitus. The beneficial effect appears to be due to inhibition of the inflammatory response to chemical irritation.

Aspiration of vomitus usually occurs during inhalation anesthesia. Obstetrical patients appear to be particularly liable. Such aspiration may be followed by the development of a clinical syndrome (Mendelson's syndrome) within two to five hours, consisting of cyanosis, dyspnea, tachycardia and shock. Physical signs may include those due to shock and those due to pulmonary edema and bronchoconstriction. X-ray of the chest may show soft patchy areas of consolidation throughout the lung fields. Deterioration of the patient's condition with a fatal termination may occur rapidly.

Treatment consists of the immediate institution of all measures necessary to oxygenate the patient and clear the airway. These include discontinuance of the general anesthetic, aspiration of vomitus from the pharynx and larynx, clearance of the larynx and bronchial tree under direct laryngoscopy and bronchoscopy, and positioning of the patient to minimize the possibility of further aspiration.

Hydrocortisone sodium succinate 100 mg should be given immediately and repeated every six to eight hours for two or three days or until the chest. The same dosage may be employed in children. Intravenous administration of the initial dose is recommended. If desired, subsequent doses may be given by intravenous infusion or intramuscularly. Full doses of a broad range antibiotic or combination of antibiotics should be given to prevent the development of secondary infection. Oxygen should be given. If bronchoconstriction is prominent, intravenous administration of a bronchodilator drug (aminophylline, isoproterenol) may be beneficial. Expectant cough mixtures may aid in the removal of bronchial secretions.

CONTRAINDICATIONS

Hydrocortisone is contra-indicated where there is known hypersensitivity to components and in systemic fungal infection unless specific anti-infective therapy is employed.

Administration of live or live, attenuated vaccines is contraindicated in patients receiving immunosuppressive doses of corticosteroids.

WARNINGS AND PRECAUTIONS

- Undesirable effects may be minimised by using the lowest effective dose for the minimum period. Frequent patient review is required to appropriately titrate the dose against disease activity.
 - Adrenal cortical atrophy develops during prolonged therapy and may persist for months after stopping treatment. In patients who have received more than physiological doses of systemic corticosteroids (approximately 30 mg hydrocortisone) for greater than 3 weeks, withdrawal should not be abrupt. How dose reduction should be carried out depends largely on whether the disease is likely to relapse as the dose of systemic corticosteroids is reduced. Clinical assessment of disease activity may be needed during withdrawal. If the disease is unlikely to relapse on withdrawal of systemic corticosteroids, but there is uncertainty about HPA suppression, the dose of systemic corticosteroid may be reduced rapidly to physiological doses. Once a daily dose of 30 mg hydrocortisone is reached, dose reduction should be slower to allow the HPA-axis to recover.
 - Abrupt withdrawal of systemic corticosteroid treatment, which has continued up to 3 weeks are appropriate if it considered that the disease is unlikely to relapse. Abrupt withdrawal of doses up to 160 mg hydrocortisone for 3 weeks is unlikely to lead to clinically relevant HPA-axis suppression, in the majority of patients. In the following patient groups, gradual withdrawal of systemic corticosteroid therapy should be considered even after courses lasting 3 weeks or less:
 - Patients who have had repeated courses of systemic corticosteroids, particularly if taken for greater than 3 weeks.
 - When a short course has been prescribed within one year of cessation of long-term therapy (months or years).
 - Patients who may have reasons for adrenocortical insufficiency other than exogenous corticosteroid therapy.
 - Patients receiving doses of systemic corticosteroid greater than 160 mg hydrocortisone.
 - Patients repeatedly taking doses in the evening.
 - Corticosteroids may mask some signs of infection, and new infections may appear during their use. Suppression of the inflammatory response and immune function increases the susceptibility to fungal, viral and bacterial infections and their severity. The clinical presentation may often be atypical and may reach an advanced stage before being recognised.
 - Chickenpox is of serious concern since this normally minor illness may be fatal in immunosuppressed patients. Patients (or parents of children) without a definite history of chickenpox should be advised to avoid close personal contact with chickenpox or herpes zoster and if exposed they should seek urgent medical attention. Passive immunization with varicella/zoster immunoglobulin (VZIG) is needed by exposed non-immune patients who are receiving systemic corticosteroids or who have used them within the previous 3 months; this should be given within 10 days of exposure to chickenpox. If a diagnosis of chickenpox is confirmed, the illness warrants specialist care and urgent treatment. Corticosteroids should not be stopped and the dose may need to be increased.
 - Exposure to measles should be avoided. Medical advice should be sought immediately if exposure occurs. Prophylaxis with normal intramuscular immunoglobulin may be needed.
 - Live vaccines should not be given to individuals with impaired immune responsiveness. The antibody response to other vaccines may be diminished.
 - The use of hydrocortisone in active tuberculosis should be restricted to those cases of fulminating or disseminated tuberculosis in which the corticosteroid is used for the management of the disease in conjunction with appropriate antituberculosis regimen. If corticosteroids are indicated in patients with latent tuberculosis or tuberculin reactivity, close observation is necessary as reactivation of the disease may occur. During prolonged corticosteroid therapy, these patients should receive chemoprophylaxis.
 - Rarely anaphylactoid reactions have been reported following parenteral hydrocortisone therapy. Physicians using the drug should be prepared to deal with such a possibility. Appropriate precautionary measures should be taken prior to administration, especially when the patient has a history of drug allergy.
 - Care should be taken for patients receiving cardioactive drugs such as digoxin because of steroid induced electrolyte disturbance/potassium loss.
 - Hydrocortisone may have an increased effect in patients with liver diseases since the metabolism and elimination of hydrocortisone is significantly decreased in these patients.
- Special precautions:** Particular care is required when considering the use of systemic corticosteroids in patients with the following conditions and frequent patient monitoring is necessary.
- Osteoporosis (post-menopausal females are particularly at risk).
 - Hypertension or congestive heart failure.
 - Existing or previous history of severe affective disorders (especially previous steroid psychosis).
 - Diabetes mellitus (or a family history of diabetes).
 - History of tuberculosis.
 - Glaucoma (or a family history of glaucoma).
 - Previous corticosteroid-induced myopathy.
 - Liver failure or cirrhosis.
 - Renal insufficiency.
 - Epilepsy.
 - Peptic ulceration.
 - Fresh intestinal anastomoses.
 - Predisposition to thrombophlebitis.
 - Abscess or other pyogenic infections.
 - Ulcerative colitis.
 - Diverticulitis.
 - Myasthenia gravis.
 - Ocular herpes simplex, for fear of corneal perforation.
 - Hypothyroidism.
 - Recent myocardial infarction (myocardial rupture has been reported).
 - Kaposi's sarcoma has been reported to occur in patients receiving corticosteroid therapy. Discontinuation of corticosteroids may result in clinical remission.
 - Pheochromocytoma crisis, which can be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after an appropriate risk/benefit evaluation.
 - Hydrocortisone can cause elevation of blood pressure, salt and water retention and increased excretion of potassium. Dietary salt restriction and potassium supplementation may be necessary. All corticosteroids increase calcium excretion.
 - Patients and/or carers should be warned that potentially severe psychiatric adverse reactions may occur with systemic steroids. Symptoms typically emerge within a few days or weeks of starting treatment. Risks may be higher with high doses/systemic exposure, although dose levels do not allow prediction of the onset, type, severity or duration of reactions. Most reactions recover after either dose reduction or withdrawal, although specific treatment may be necessary. Patients/carers should be encouraged to seek medical advice if worrying psychological symptoms develop, especially if depressed mood or suicidal ideation is suspected.

hovid Hydrocortisone Sodium Succinate for Injection 100 mg

Patients/carers should be alert to possible psychiatric disturbances that may occur either during or immediately after dose tapering/withdrawal of systemic steroids, although such reactions have been reported infrequently.

Particular care is required when considering the use of systemic corticosteroids in patients with existing or previous history of severe affective disorders in themselves or in their first degree relatives. These would include depressive or manic-depressive illness and previous steroid psychosis.

Use in children: Corticosteroids cause growth retardation in infancy, childhood and adolescence, which may be irreversible. Treatment should be limited to the minimum dosage for the shortest possible time. The use of steroids should be restricted to the most serious indications.

Use in the elderly: The common adverse effects of systemic corticosteroids may be associated with more serious consequences in old age, especially osteoporosis, hypertension, hypokalaemia, diabetes, susceptibility to infection and thinning of the skin. Close clinical supervision is required to avoid life-threatening reactions.

"Corticosteroids should not be used for the management of head injury or stroke because it is unlikely to be of benefit and may even be harmful."

Effects on ability to drive and use machines: The effect of corticosteroids on the ability to drive or use machinery has not been systematically evaluated. Undesirable effects, such as syncope, vertigo, and convulsions are possible after treatment with corticosteroids. If affected, patients should not drive or operate machinery.

PREGNANCY AND LACTATION

Pregnancy: The ability of corticosteroids to cross the placenta varies between individual drugs, however, hydrocortisone readily crosses the placenta.

Administration of corticosteroids to pregnant animals can cause abnormalities of foetal development including cleft palate, intra-uterine growth retardation and affects on brain growth and development. There is no evidence that corticosteroids result in an increased incidence of congenital abnormalities, such as cleft palate in man, however, when administered for long periods or repeatedly during pregnancy, corticosteroids may increase the risk of intra-uterine growth retardation. Hypoadrenalism may, in theory, occur in the neonate following prenatal exposure to corticosteroids but usually resolves spontaneously following birth and is rarely clinically important. As with all drugs, corticosteroids should only be prescribed when the benefits to the mother and child outweigh the risks. When corticosteroids are essential, however, patients with normal pregnancies may be treated as though they were in the non-gravid state.

Lactation: Corticosteroids are excreted in breast milk, although no data are available for hydrocortisone. Doses up to 160 mg daily of hydrocortisone are unlikely to cause systemic effects in the infant. Infants of mothers taking higher doses than this may have a degree of adrenal suppression, but the benefits of breastfeeding are likely to outweigh any theoretical risk.

DRUG INTERACTIONS

- Convulsions have been reported with concurrent use of corticosteroids and cyclosporin. Since concurrent administration of these agents results in a mutual inhibition of metabolism, it is possible that convulsions and other adverse effects associated with the individual use of either drug may be more apt to occur.
- Drugs that induce hepatic enzymes, such as rifampicin, rifabutin, carbamazepine, phenobarbitone, phenytoin, primidone, and aminoglutethimide enhance the metabolism of corticosteroids and its therapeutic effects may be reduced.
- Drugs which inhibit the CYP3A4 enzyme, such as cimetidine, erythromycin, ketoconazole, itraconazole, diltiazem and mibefradil, may decrease the rate of metabolism of corticosteroids and hence increase the serum concentration.
- Steroids may reduce the effects of anticholinesterases in myasthenia gravis. The desired effects of hypoglycaemic agents (including insulin), anti-hypertensives and diuretics are antagonised by corticosteroids, and the hypokalaemic effects of acetazolamide, loop diuretics, thiazide diuretics and carbenoxolone are enhanced.
- The efficacy of coumarin anticoagulants may be enhanced by concurrent corticosteroid therapy and close monitoring of the INR or prothrombin time is required to avoid spontaneous bleeding.
- The renal clearance of salicylates is increased by corticosteroids and steroid withdrawal may result in salicylate intoxication. Salicylates and non-steroidal anti-inflammatory agents should be used cautiously in conjunction with corticosteroids in hypothermiaemia.
- Steroids have been reported to interact with neuromuscular blocking agents such as pancuronium with partial reversal of the neuromuscular block.

MAIN SIDE/ADVERSE EFFECTS

Hydrocortisone is normally employed on a short-term basis it is unlikely that side effects will occur; however, the possibility of side effects attributable to corticosteroid therapy should be recognized. Such side effects include:

Adverse Effects Table	
System Organ Class	Frequency Not Known (Cannot be estimated from available data)
Infections and infestations	Infection masked; Opportunistic infection
Neoplasms benign, malignant and unspecified (including cysts and polyps)	Kaposi's sarcoma (has been reported to occur in patients receiving corticosteroid therapy)
Immune system disorders	Hypersensitivity (including anaphylaxis and anaphylactoid reactions [e.g. bronchospasm, laryngeal oedema, urticaria]); May suppress reactions to skin tests
Blood and lymphatic system disorders	Leucocytosis
Endocrine disorders	Cushingoid; Pituitary-adrenal axis suppression; WITHDRAWAL SYMPTOMS - Too rapid a reduction of corticosteroid dosage following prolonged treatment can lead to acute adrenal insufficiency, hypotension and death. However, this is more applicable to corticosteroids with an indication where continuous therapy is given (see section 4.4); A 'withdrawal syndrome' may also occur including, fever, myalgia, arthralgia, rhinitis, conjunctivitis, painful itchy skin nodules and loss of weight
Metabolism and nutrition disorders	Sodium retention; Water retention; Alkalosis hypokalaemic; Glucose tolerance impaired; Increased appetite; Weight increased
Psychiatric disorders	Affective disorders (such as irritable, euphoric, depressed and labile mood psychological dependence and suicidal thoughts); Psychotic reactions (including mania, delusions, hallucinations and aggravation of schizophrenia); Behavioural disturbances; Irritability; Anxiety; Sleep disturbances; Cognitive dysfunction including confusion and amnesia

Nervous system disorders	Increased intra-cranial pressure with papilloedema in children (pseudotumour cerebri) has been reported, usually after treatment withdrawal of hydrocortisone; Benign intracranial hypertension; Convulsions; Epidural lipomatosis
Eye disorders	Cataract subcapsular; Glaucoma; Exophthalmos; Increased intra-ocular pressure, with possible damage to the optic nerve; Corneal or scleral thinning; Exacerbation of ophthalmic viral or fungal disease, Central serous chorioretinopathy
Cardiac disorders	Cardiac failure congestive (in susceptible patients); Myocardial rupture following a myocardial infarction
Vascular disorders	Hypertension; Thrombosis including Thromboembolism
Respiratory, thoracic and mediastinal disorders	Hiccups; Pulmonary embolism
Gastrointestinal disorders	Peptic ulcer (with possible perforation and haemorrhage); Gastric haemorrhage; Pancreatitis; Abdominal distension; Oesophageal ulceration; Oesophageal candidiasis; Intestinal perforation; Dyspepsia; Nausea
Skin & subcutaneous tissue disorders	Petechiae; Telangiectasia; Eczymosis; Skin atrophy; Skin striae; Skin hyperpigmentation; Skin hypopigmentation; Hirsutism; Acne; Hyperhidrosis
Musculoskeletal, connective tissue and bone disorders	Myopathy; Muscular weakness; Osteonecrosis; Osteoporosis; Pathological fracture; Growth retardation
Reproductive system and breast disorders	Menstruation irregular; Amenorrhoea
General disorders and administration site conditions	Impaired healing; Abscess sterile; Malaise
Investigations	Carbohydrate tolerance decreased; Increased insulin requirement (or oral hypoglycaemic agents in diabetics); Blood potassium decreased; Nitrogen balance negative (due to protein catabolism); Urine calcium increased; Alanine aminotransferase increased; Aspartate aminotransferase increased; Blood alkaline phosphatase increased
Injury, poisoning and procedural complications	Spinal compression fracture; Tendon rupture (particularly of the Achilles tendon)

OVERDOSE AND TREATMENT

There is no clinical syndrome of acute overdosage with hydrocortisone. Hydrocortisone is dialysable.

DOSAGE AND ADMINISTRATION

Because of possible physical incompatibilities, hydrocortisone sodium succinate should not be diluted or mixed with other solutions.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

This preparation may be administered by intravenous injection, by intravenous infusion, or by intra muscular injection, the preferred method for initial emergency use being intravenous injection. Following the initial emergency period, consideration should be given to employing a longer acting injectable preparation or an oral preparation.

Therapy is initiated by administering hydrocortisone sodium succinate intravenously over a period of 30 seconds (eg, 100 mg) to 10 minutes (e.g., 500 mg or more). In general, high dose corticosteroid therapy should be continued only until the patient's condition has stabilized - usually not beyond 48 to 72 hours. When high dose hydrocortisone therapy must be continued beyond 48–72 hours, hyponatremia may occur. Under such circumstances it may be desirable to replace hydrocortisone sodium succinate with a corticoid such as methylprednisolone sodium succinate which causes little or no sodium retention. The initial dose of hydrocortisone sodium succinate is 100 mg to 500 mg, depending on the specific disease entity being treated. However, in certain overwhelming, acute, life-threatening situations, administration in dosages exceeding the usual dosages may be justified and may be in multiples of the oral dosages.

This dose may be repeated at intervals of 2, 4 or 6 hours as indicated by the patient's response and clinical condition.

It should be emphasized that dosage requirements are variable and must be individualized on the basis of the disease under treatment and the response of the patient. After a favorable response is noted, the proper maintenance dosage should be determined by decreasing the initial drug dosage in small decrements at appropriate time intervals until the lowest dosage that maintains an adequate clinical response is reached. Situations which may make dosage adjustments necessary are changes in clinical status secondary to remissions or exacerbations in the disease process, the patient's individual drug responsiveness, and the effect of patient exposure to stressful situations not directly related to the disease entity under treatment. In this latter situation it may be necessary to increase the dosage of the corticosteroid for a period of time consistent with the patient's condition. If after long-term therapy the drug is to be stopped, it is recommended that it be withdrawn gradually rather than abruptly.

In patients with liver disease, there may be an increased effect and reduced dosing may be considered.

In pediatric patients, the initial dose of hydrocortisone may vary depending on the specific disease entity being treated. The range of initial doses is 0.56 to 8 mg/kg/day in three or four divided doses (20 to 240 mg/m²/day). For the purpose of comparison, the following is the equivalent milligram dosage of the various glucocorticoids:

Cortisone, 25	Triamcinolone, 4
Hydrocortisone, 20	Paramethasone, 2
Prednisolone, 5	Betamethasone, 0.75
Prednisone, 5	Dexamethasone, 0.75
Methylprednisolone, 4	

These dose relationships apply only to oral or intravenous administration of these compounds. When these substances or their derivatives intramuscularly or into joint spaces, their relative properties may be greatly altered.

Geriatric Use: In general, dose selection for an elderly patient should be cautious, usually starting at the low end of the dosing range, reflecting the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy.

Note: The information given here is limited. For further information consult your doctor or pharmacist.

Storage: Store below 30°C. Protect from light and moisture.

Presentation/ Packing: The vials are packed in boxes of 1, 10 and 50. (Not all presentations are available locally)

Product Registration Holder/Repacked by: **HOVID Bhd.**
121, Jalan Tunku Abdul Rahman, 30010 Ipoh, Malaysia.
Manufactured by: Tianjin Biochemical Pharmaceutical Co., Ltd.,
269, Huanhuan Road, Tianjin Airport Economical Area, Tianjin 300308, China.

Revision date: September 2017